

A Non-Recursive Composition Formula for Inverse-Dyadic Values of the Fabius Function

Abstract

Let $F_n = F(2^{-n})$ denote the inverse-dyadic values of the Fabius function. Starting from the triangular recurrence

$$F_n = \frac{2^{-\binom{n}{2}}}{2^n - 1} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}}}{(n-k+1)!} F_k, \quad n \geq 1,$$

with $F_0 = 1$, we derive a finite, fully non-recursive solution. After the normalization $a_n = 2^{\binom{n}{2}} F_n$, every term becomes the total weight of all increasing paths from 0 to n . Equivalently, it is a sum over the ordered compositions of n . We prove the formula by decomposing a composition at its last block, give equivalent increasing-chain and nilpotent-matrix forms, and verify the values through $n = 4$. The argument is purely finite: no limiting process or convergence theorem is involved. We also formalize the path and composition arguments in Lean.

1 The recurrence and its range

We study a sequence $(F_n)_{n \geq 0}$ with initial value

$$F_0 = 1 \tag{1}$$

and recurrence

$$F_n = \frac{2^{-\binom{n}{2}}}{2^n - 1} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}}}{(n-k+1)!} F_k, \quad n \geq 1. \tag{2}$$

For the Fabius function, $F_n = F(2^{-n})$, so that $F_0 = F(1) = 1$. Everything below is algebraic and applies to any sequence satisfying (1)–(2).

The restriction $n \geq 1$ is essential. At $n = 0$ the denominator $2^n - 1$ vanishes and the displayed sum is empty. Thus (2) is not a formula for F_0 ; the initial value (1) must be supplied separately. In a formal field such as Lean’s rational or real numbers, division by zero is totalized, but that convention does not repair the identity: its right-hand side would reduce to zero while $F_0 = 1$.

For every $n \geq 1$, all quantities in (2) are defined and its right-hand side involves only earlier values $F_0, \dots, F_{n-1}$. It therefore determines the sequence uniquely. Our aim is to replace this recursive determination by one finite multiple sum containing no F_k .

2 Normalization

The triangular powers of two in (2) are designed to telescope. This is clearest after a normalization.

Definition 2.1. For $n \geq 0$, put

$$a_n = 2^{\binom{n}{2}} F_n. \tag{3}$$

Since $\binom{0}{2} = 0$, one has $a_0 = 1$. Multiplying (2) by $2^{\binom{n}{2}}$ and replacing $2^{\binom{k}{2}}F_k$ by a_k gives

$$a_n = \frac{1}{2^n - 1} \sum_{k=0}^{n-1} \frac{a_k}{(n-k+1)!}, \quad n \geq 1. \quad (4)$$

Equivalently, if

$$T_{n,k} = \frac{1}{(2^n - 1)(n-k+1)!} \quad (0 \leq k < n), \quad (5)$$

then

$$a_n = \sum_{k=0}^{n-1} T_{n,k} a_k. \quad (6)$$

The normalized recurrence is a weighted path recurrence on the acyclic directed graph whose vertices are the nonnegative integers and which has an edge $k \rightarrow n$ whenever $k < n$.

Proposition 2.2 (Triangular uniqueness). *There is at most one sequence over $\mathbb{Q}$ or $\mathbb{R}$ satisfying $a_0 = 1$ and (4). Consequently there is at most one sequence satisfying (1) and (2).*

Proof. If two normalized sequences agree through index $n-1$, then (4) expresses both values at index n as the same linear combination of those earlier values. Induction proves equality at every index. The change of variables (3) is invertible, so the same conclusion holds for (F_n) . $\square$

3 The composition formula

An *ordered composition* of n is a finite tuple of positive integers whose sum is n . We write

$$\rho = (r_1, \dots, r_m) \models n$$

and call $m = \text{len}(\rho)$ its length. Let $\mathcal{C}(n)$ be the set of all such compositions. We use the standard convention

$$\mathcal{C}(0) = \{()\},$$

where $()$ is the empty composition. For a nonempty composition define its partial sums

$$s_j = r_1 + \dots + r_j, \quad 1 \leq j \leq m. \quad (7)$$

Thus $s_m = n$. Define the weight

$$\text{wt}(\rho) = \prod_{j=1}^m \frac{1}{(2^{s_j} - 1)(r_j + 1)!}. \quad (8)$$

For the empty composition the product is empty and has value 1.

Theorem 3.1 (Non-recursive composition formula). *For every $n \geq 0$,*

$$F_n = 2^{-\binom{n}{2}} \sum_{\rho \in \mathcal{C}(n)} \prod_{j=1}^{\text{len}(\rho)} \frac{1}{(2^{s_j} - 1)(r_j + 1)!} \quad (9)$$

where $\rho = (r_1, \dots, r_m)$ and s_j is given by (7). At $n = 0$, the right-hand side is interpreted using the unique empty composition and equals 1.

Proof. Let

$$A_n = \sum_{\rho \in \mathcal{C}(n)} \text{wt}(\rho).$$

The empty-composition convention gives $A_0 = 1$. Fix $n \geq 1$. Every composition $\rho = (r_1, \dots, r_m)$ of n has a unique last block. Put

$$k = r_1 + \dots + r_{m-1},$$

where $k = 0$ if $m = 1$. Then $0 \leq k < n$, the initial tuple $(r_1, \dots, r_{m-1})$ is a composition of k , and the last block is $r_m = n - k$. Conversely, appending $n - k$ to any composition of k produces a composition of n with penultimate partial sum k .

The last factor in (8) is

$$\frac{1}{(2^n - 1)(n - k + 1)!} = T_{n,k}.$$

All earlier factors form precisely the weight of the prefix composition. Partitioning $\mathcal{C}(n)$ according to k therefore gives

$$A_n = \sum_{k=0}^{n-1} \frac{A_k}{(2^n - 1)(n - k + 1)!} = \frac{1}{2^n - 1} \sum_{k=0}^{n-1} \frac{A_k}{(n - k + 1)!}. \quad (10)$$

Thus (A_n) satisfies the same recurrence and initial value as (a_n) . By Proposition 2.2, $A_n = a_n$ for every n . Finally, $F_n = 2^{-\binom{n}{2}} a_n$ by (3), proving (9). $\square$

Remark 3.2 (Finiteness). For $n \geq 1$, there are 2^{n-1} ordered compositions of n . Hence (9) is a finite sum of 2^{n-1} positive rational terms. It is fully non-recursive, although its number of terms grows exponentially. No convergence question is present.

Corollary 3.3 (Positivity). $F_n > 0$ for every $n \geq 0$.

Proof. The $n = 0$ case is immediate. If $n \geq 1$, then every partial sum s_j in a composition is positive, so $2^{s_j} - 1 > 0$, and every factorial is positive. Every summand in (9) is therefore positive. $\square$

4 Increasing chains and an explicit multiple sum

The composition formula can be written as a path sum without introducing blocks. For $m \geq 1$, let $\mathcal{P}_m(n)$ denote the set of strictly increasing chains

$$0 = i_0 < i_1 < \dots < i_m = n. \quad (11)$$

The bijection between compositions and chains is

$$r_j = i_j - i_{j-1}, \quad i_j = r_1 + \dots + r_j = s_j. \quad (12)$$

Under this bijection, (8) becomes

$$\prod_{j=1}^m \frac{1}{(2^{i_j} - 1)(i_j - i_{j-1} + 1)!}. \quad (13)$$

Corollary 4.1 (Increasing-chain form). For $n \geq 1$,

$$F_n = 2^{-\binom{n}{2}} \sum_{m=1}^n \sum_{0=i_0 < i_1 < \dots < i_m=n} \prod_{j=1}^m \frac{1}{(2^{i_j} - 1)(i_j - i_{j-1} + 1)!}. \quad (14)$$

Formula (14) is the requested multiple sum. Written with only the internal indices displayed, its inner summation is

$$\sum_{1 \leq i_1 < \dots < i_{m-1} \leq n-1} \prod_{j=1}^m \frac{1}{(2^{i_j} - 1)(i_j - i_{j-1} + 1)!}, \quad i_0 = 0, \quad i_m = n.$$

When $m = 1$, there are no internal indices and the inner sum consists of the single chain $0 < n$.

There is also a direct explanation for the common factor $2^{-\binom{n}{2}}$. Before normalization, the edge $k \rightarrow r$ in (2) has weight

$$U_{r,k} = \frac{2^{\binom{k}{2} - \binom{r}{2}}}{(2^r - 1)(r - k + 1)!}. \quad (15)$$

Along a chain (11), the powers of two telescope:

$$\begin{aligned} \prod_{j=1}^m 2^{\binom{i_{j-1}}{2} - \binom{i_j}{2}} &= 2^{\sum_{j=1}^m (\binom{i_{j-1}}{2} - \binom{i_j}{2})} \\ &= 2^{\binom{i_0}{2} - \binom{i_m}{2}} = 2^{-\binom{n}{2}}. \end{aligned}$$

The remaining edge factors are exactly (13).

5 A nilpotent-matrix interpretation

Fix a terminal index N and consider matrices indexed by $\{0, 1, \dots, N\}$. Define the strictly lower-triangular matrix

$$T_{i,j}^{[N]} = \begin{cases} \frac{1}{(2^i - 1)(i - j + 1)!}, & 0 \leq j < i \leq N, \\ 0, & j \geq i. \end{cases} \quad (16)$$

Let

$$\mathbf{a}^{[N]} = (a_0, a_1, \dots, a_N)^\top, \quad \mathbf{e}_0 = (1, 0, \dots, 0)^\top.$$

The initial condition and (6) say

$$\mathbf{a}^{[N]} = \mathbf{e}_0 + T^{[N]} \mathbf{a}^{[N]}. \quad (17)$$

Because $T^{[N]}$ is strictly lower triangular, $(T^{[N]})^{N+1} = 0$. Therefore

$$(I - T^{[N]})^{-1} = I + T^{[N]} + \dots + (T^{[N]})^N, \quad (18)$$

and (17) yields

$$\mathbf{a}^{[N]} = \left(I + T^{[N]} + \dots + (T^{[N]})^N \right) \mathbf{e}_0. \quad (19)$$

Taking the n th component, for any $n \leq N$, gives

$$F_n = 2^{-\binom{n}{2}} \sum_{m=0}^n (T^{[N]})_{n,0}^m. \quad (20)$$

For $n > 0$, the $m = 0$ term vanishes. Standard matrix multiplication shows that $(T^{[N]})_{n,0}^m$ is the sum of the products of edge weights over all length- m increasing paths from 0 to n . Thus (20) is precisely Corollary 4.1.

If $D = \text{diag}(2^{\binom{0}{2}}, \dots, 2^{\binom{N}{2}})$, then the matrix acting directly on $(F_0, \dots, F_N)^\top$ is $U = D^{-1} T^{[N]} D$. Its entries are (15); the diagonal similarity explains the telescoping powers along paths.

6 Low-index examples

We now expand the composition formula explicitly. These checks both illustrate the weighting rule and guard against shifts in the factorial or the Mersenne factor.

$n = 0$ **and** $n = 1$

For $n = 0$, the sole composition is empty, so $a_0 = 1$ and $F_0 = 1$. For $n = 1$, the sole composition is (1), with weight

$$\frac{1}{(2^1 - 1)(1 + 1)!} = \frac{1}{2}.$$

Since $2^{-\binom{1}{2}} = 1$, this gives

$$F_1 = \frac{1}{2}.$$

$n = 2$

The compositions are (2) and (1, 1). Their weights are

$$\text{wt}(2) = \frac{1}{(2^2 - 1)3!} = \frac{1}{18}, \quad \text{wt}(1, 1) = \frac{1}{(2^1 - 1)2!} \frac{1}{(2^2 - 1)2!} = \frac{1}{12}.$$

Hence

$$a_2 = \frac{1}{18} + \frac{1}{12} = \frac{5}{36}, \quad F_2 = 2^{-1}a_2 = \frac{5}{72}.$$

$n = 3$

For (3), (1, 2), (2, 1), and (1, 1, 1) the respective weights are

$$\frac{1}{168}, \quad \frac{1}{84}, \quad \frac{1}{252}, \quad \frac{1}{168}.$$

Thus

$$a_3 = \frac{1}{168} + \frac{1}{84} + \frac{1}{252} + \frac{1}{168} = \frac{1}{36}, \quad F_3 = 2^{-3}a_3 = \frac{1}{288}.$$

$n = 4$

There are eight compositions. In the order

$$(4), (1, 3), (2, 2), (3, 1), (1, 1, 2), (1, 2, 1), (2, 1, 1), (1, 1, 1, 1),$$

their weights are

$$\frac{1}{1800}, \quad \frac{1}{720}, \quad \frac{1}{1620}, \quad \frac{1}{5040}, \quad \frac{1}{1080}, \quad \frac{1}{2520}, \quad \frac{1}{7560}, \quad \frac{1}{5040}.$$

Their sum and the resulting inverse-dyadic value are

$$a_4 = \frac{143}{32400}, \quad F_4 = 2^{-6}a_4 = \frac{143}{2073600}.$$

For reference, the first five values are therefore

n	$a_n = 2^{\binom{n}{2}} F_n$	F_n
0	1	1
1	1/2	1/2
2	5/36	5/72
3	1/36	1/288
4	143/32400	143/2073600

When $F_n = F(2^{-n})$, this reads

$$F(1) = 1, \quad F(1/2) = \frac{1}{2}, \quad F(1/4) = \frac{5}{72}, \quad F(1/8) = \frac{1}{288}, \quad F(1/16) = \frac{143}{2073600}.$$

7 Equivalence of the three viewpoints

We summarize the logical equivalence, which is useful when choosing a form for computation or formal proof.

Proposition 7.1. *For a sequence with $F_0 = 1$, the following descriptions are equivalent.*

1. *The sequence satisfies the recurrence (2) for every $n \geq 1$.*
2. *It satisfies the composition formula (9) for every $n \geq 0$.*
3. *Its normalization satisfies the finite matrix formula (19) for every terminal index N .*

Proof. The implication from the composition formula to the normalized recurrence is the last-block decomposition (10); undoing the normalization gives (2). Conversely, the recurrence and Proposition 2.2 identify its solution with the composition sum constructed in Theorem 3.1. Finally, (17) is simply the normalized recurrence written for all indices up to N , and the finite geometric identity (18) is equivalent to its unique solution. Expanding matrix powers gives the chain formula, which is carried to the composition formula by the bijection (12). $\square$

This proposition also clarifies the meaning of “closed form” here. The composition expression is not recursive: each summand depends only on the blocks of one composition and on n . The matrix inverse is likewise a finite polynomial in a nilpotent matrix. Both are exact finite solutions of the triangular system.

8 Relation to inverse-dyadic Fabius values

The recurrence arises naturally from the half moments of Rvachev’s up function. If d_n denotes the n th half moment and

$$a_n = \frac{d_n}{n!},$$

then the moment recurrence gives (4). The analytic bridge to the Fabius function is

$$d_n = n! 2^{\binom{n}{2}} F(2^{-n}),$$

so the two definitions of a_n agree. Substituting this bridge into (4) yields (2), and Theorem 3.1 therefore supplies a non-recursive exact formula for every inverse-dyadic Fabius value.

There is no ambiguity between the usual bounded Fabius function and its signed global extension in this statement. Every argument 2^{-k} belongs to $[0, 1]$, including $2^0 = 1$, and the two versions agree throughout that interval. Thus the same finite composition formula applies to either notation.

9 Lean formalization

The argument above has been formalized in Lean. The implementation is in

`FabiusFunction/FabiusInverseDyadicClosedForm.lean`

and is exported by the public `FabiusFunction` facade. It contains no axioms or placeholders beyond Lean and Mathlib’s standard logical foundations. The proof is split into a reusable path-sum layer, the explicit composition weights, and the analytic Fabius corollaries.

Existing recurrence layer

The pre-existing normalized sequence is represented by

`fabiusRecurrenceSequence n = halfMoment n / n!`

with the positive-index recurrence exposed as

`fabiusRecurrenceSequence_recurrence`
`(n : Nat) (hn : 1 <= n) : ...`

The exact inverse-dyadic bridge and direct recurrence are represented by declarations such as

`fabius_inverse_two_pow_eq_recurrenceSequence`
`fabiusAtInverseTwoPow_recurrence_zpow`
`fabiusFunction_inverse_two_pow_recurrence_zpow`
`globalFabius_inverse_two_pow_recurrence`

The hypothesis $1 \leq n$ remains explicit in source-facing recurrence theorems. The non-recursive formula itself is total at $n = 0$, because it uses the empty composition rather than the singular recurrence.

Generic weighted paths

Mathlib provides the type `Composition n` in `Mathlib.Combinatorics.Enumerative.Composition`. Its `blocks` list records the positive increments of a path. The formal equivalence

`Composition.splitLastEquiv`

identifies a composition of a positive n with a proper predecessor $k < n$ and a composition of k . The definitions `Composition.pathWeight` and `Composition.pathSum` multiply edge weights and sum them over all compositions. The last-edge decomposition is the theorem

`Composition.pathSum_eq_sum_range`

and the general solver

`triangularRecurrence_eq_initial_mul_pathSum`

proves, by strong induction, that any semiring-valued triangular recurrence is its initial value times the corresponding finite path sum. A separate `triangularRecurrence_unique` theorem records uniqueness.

Explicit composition formula

The theorem

`Composition.pathWeight_eq_boundary_product`

rewrites a recursive list product chronologically, between consecutive partial sums. Since `blocksFun` gives the blocks r_j and `sizeUpTo` gives the partial sums s_j , the article's rational weight is implemented literally as

```
def fabiusCompositionWeight (c : Composition n) : Rat :=  
  product over j of  
    1 / ((2^(c.sizeUpTo (j+1))) - 1) * (c.blocksFun j + 1)!
```

and `fabiusCompositionSum n` sums this weight over all `Composition n`. The principal exact statements are

```
fabiusCompositionSum_eq_pathSum  
fabiusRecurrenceSequence_eq_sum_compositions  
fabiusAtInverseTwoPow_eq_sum_compositions  
fabiusAtInverseTwoPow_eq_composition_formula  
fabiusAtInverseTwoPow_eq_composition_formula_by_length  
fabiusFunction_inverse_two_pow_eq_sum_compositions  
fabius_inverse_two_pow_eq_sum_compositions_zpow  
globalFabius_inverse_two_pow_eq_sum_compositions
```

The first fully expanded theorem is precisely (9). The second is the requested multiple-sum presentation: it first sums over the length $1 \leq m \leq n$, then over all compositions of n , retaining only those of length m . No infinite sum or matrix library is needed. The generic real theorem holds for every bounded function satisfying the Fabius equations; the canonical and signed-global versions follow from the already proved analytic bridges.

Verification targets

The mechanized theorem is valid for every $n : \mathbb{N}$, including $n = 0$. Exact rational evaluations through $n = 4$ reproduce the values in Section 6. The source has been checked directly and through the public facade; the principal declarations have no `sorry` and use only the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. The real corollaries are derived from the rational identity by the existing recurrence-sequence bridge, rather than by reproving any analytic property of the Fabius function.

10 Conclusion

The normalization $a_n = 2^{\binom{n}{2}} F_n$ turns the inverse-dyadic Fabius recurrence into a weighted acyclic path recurrence. Summing all paths gives the exact composition formula (9). Its proof is the elementary operation of removing the last block of a composition; its matrix counterpart is the finite Neumann series of a nilpotent lower-triangular matrix. The resulting expression is explicit, positive, rational, and fully non-recursive, and it incorporates the initial value at $n = 0$ through the empty-composition convention.